Rose Graves
(239)887-0963 | Houston, TX | rosekgraves@gmail.com | www.linkedin.com/in/rose-graves

Education

Ph.D. **Statistics, August 2022-Present,** Rice University
Advisor: Prof. Marina Vannucci, Prof. Scott Powers
Thesis Committee: Prof. Kathy Ensor, Prof. Amanda Glazer

B.S. **Statistical Science, May 2020**, Duke University
Minors: Mathematics, French
Thesis: "Corrupted Data and the Illicit Arms Trade"
Advisor: Prof Alexander Volfovsky

Research Interests

Theory and Methods: Bayesian Analysis, Social Network Analysis

Applications: Sports Analytics, Political Security, Environmental Protection

Awards

- Ken Kennedy Recruitment Fellowship, Rice University
- Graduation with Highest Distinction, Duke University
- Forever Duke Student Leadership Award, Duke University
- Dean's List 2016 & 2017, Duke University

Publications

1. **Graves, R.,** Kowal, D., Vannucci, M. (2025), "Bayesian Multivariate Rank Regression Models for the Analysis of Sports Data". *submitted.*

2. Sérgio Boggio, P., Gaudêncio Rêgo, G., AC Everett, J., Graziela Bonato Vieira, **Graves, R.,** Sinnott-Armstrong, W. (2023), "Who did it? Moral wrongness for us and them in the UK, US, and Brazil", Philosophical Philosophy, pages 1-21, https://doi.org/10.1080/09515089

3. McDonald, K.*, **Graves, R.*,** Yin, S., Weese, T., and Sinnott-Armstrong, W. (2021), "Valence Framing Effects on Moral Judgements: A Meta Analysis", Cognition, 212, 104703.
   * Co-first author

4. Nadelhoffer, T., **Graves, R.,** Skorburg, G., Leary, M., Sinnott-Armstrong, W. (2020), "Partisanship, humility, and Epistemic Polarization", Polarization, Arrogance, and Dogmatism: Philosophical Perspectives (1st ed.), London, UK: Routledge.

5. Nadelhoffer, T., Yin, S., and **Graves, R.** (2020), "Folk Intuitions and the Conditional Ability to Do Otherwise", Philosophical Psychology, 33, 968-996.

Presentations & Conferences

- Conference for Texas Statisticians, Invited Talk (2026) "Bayesian Hierarchical Change Point Models for Pitcher Biomechanics"

- Symposium on Data Science and Statistics, Talk (2025) "Bayesian Multivariate Rank Regression Models for the Analysis of Sports Data"

- Conference for Texas Statisticians, Poster (2025) "Bayesian Multivariate Rank Regression Models for the Analysis of Sports Data"

Experience

- San Francisco Giants Baseball Associate Analyst | Research and Development | *February 2023 – Present*
    - Uses Bayesian Statistics methods to solve problems in player evaluation, player development, baseball strategy, and health.
    - Collaborates with the Amateur Scouting department to improve both evaluation of amateur players and draft strategy both before and during the draft.
    - Improvements in the modeling framework for evaluating player defense.
    - Created new model to evaluate player availability multiple years out.
    - Provides explanations of complex models and analyses by delivering visuals and graphics to aid in increased understanding.

- Houston Astros Baseball Operations Bayesian Analyst | Research and Development | *June 2020 – August 2022*
    - Created a new model to evaluate high school and college pitchers and batters for the 2021 & 2022 MLB draft.
    - Created and implemented novel Bayesian methods to aggregate scouting data.
    - Overhauled methods of predicting players' future injuries.
    - Improved outfield and infield positioning model.
    - Collaborated with departments across the organization including scouts and coaches.

- *Lab Manager,* Moral Attitudes and Decision-Making Lab | Duke University | *July 2019-May 2020*
    - Organized and led weekly meetings to discuss current philosophical research, plan and implement experimental philosophy research studies, and give feedback to current members' work.
    - Prepared and submitted experiment approval documents to the Institutional Review Board.
    - Provided statistics and data analysis consulting to graduate students and faculty.

International Experience

- Political Studies Institute: Strasbourg, France | *Fall 2018*
    - Completed rigorous political science courses in French.

    - Gained a global perspective on international politics and conflicts.

- National Territorial Studies Intern | Institut National des Études Territoriales (INET) | Strasbourg, France | *Fall 2018*
    - Improved the Institute's interactive educational website that aims to help educate French regional authorities.
    - Made the statistics web page more comprehensive by carrying out exploratory data analysis on data regarding local authorities in Europe.
    - Translated educational documents from French to English and helped create a glossary of industry specific terms in both languages.

Mentoring & Outreach

- American Statistical Association Rice University Chapter, Co-President, *August 2025- Present*
    - Serves as one of the founding presidents for the Rice University chapter of the American Statistical Association.
    - Organizes events and initiatives catering to the needs of students interested in statistics with aims to create and inclusive welcoming environment for all students in the statistics department.

- Graduate Student Mentor for Statistics PhD Students, *October 2025-Present*
    - Underwent mentor training to serve as a point person for two assigned mentees within the statistics department.
    - Has monthly in person chats with mentees to answer any questions about PhD life or address any concerns.

- GSA Teaching Committee Representative, GSA Representative, *August 2025- Present*
    - Advocates for improved support on teaching and resources for instructors and teaching assistants.

- American Soccer Insights Conference at Rice University, Organizer, *October 2024-Present*
    - The American Soccer Insights Conference had its inaugural conference in January 2025. With participants from the Premier League, NWSL, MLS, and many other league across the globe.
    - Assisted in the organization of the Women in Soccer Networking Event to celebrate the accomplishments of women in the field soccer analytics and provide a welcoming and inclusive environment for networking.

- *Coordinator*, Research Experience for Undergraduates, Rice University Statistics, *May 2023- August 2023*
    - Held weekly conversations between faculty and students, and bi-weekly meetings for students to present their work and receive feedback amongst a small group.
    - Assisted with any concerns and questions regarding graduate student applications, life, classes, and research.
    - Organized all social activities and events for the undergraduate researchers and faculty mentors.

- Duke Statistical Science Majors Union President, *2019-2020*

- Revived the Statistical Science Majors Union after months of inactivity and positioned it for future students to grow the organization.
- Served as the President, held weekly group meetings, organized events, moderated panels, and provided opportunities for collaborations between students and faculty.

Teaching

- *Instructor of Record,* Probability and Statistics (STAT 310), Rice University*, Fall 2025, Spring 2026.*

- *Instructor of Record,* Sports Management (SMGT 490), Rice University*, Spring 2025.* Guided students in developing questions and creating statistical analysis around a semester long sports analytics project.

- *Teaching Assistant,* Probability and Statistics (STAT 310), Rice University*, Fall 2023, Spring 2025.* Composed problem set solutions; held weekly office hours.

- *Teaching Assistant,* Linear Regression (STAT 615), Rice University*, Fall 2023.* Composed problem set solutions; held weekly office hours.

- *Teaching Assistant,* Senior Project Seminar, Statistics (STAT 450), Rice University*, Fall 2022, Spring 2023.* Gave individualized feedback on student projects on a biweekly basis; presented on effective ways of presenting, writing, and displaying results.

- *Teaching Assistant,* Introductory Statistics (Stat 101), Duke University*, Spring 2019, Fall 2020, Spring 2020.* Led Weekly labs to teach 30 students how to code in R and apply concepts learned in class.

Related Courses & Experience

- Coding Languages
  - R, Stan, SQL, Python
- Languages
  - English, French (professionally proficient)